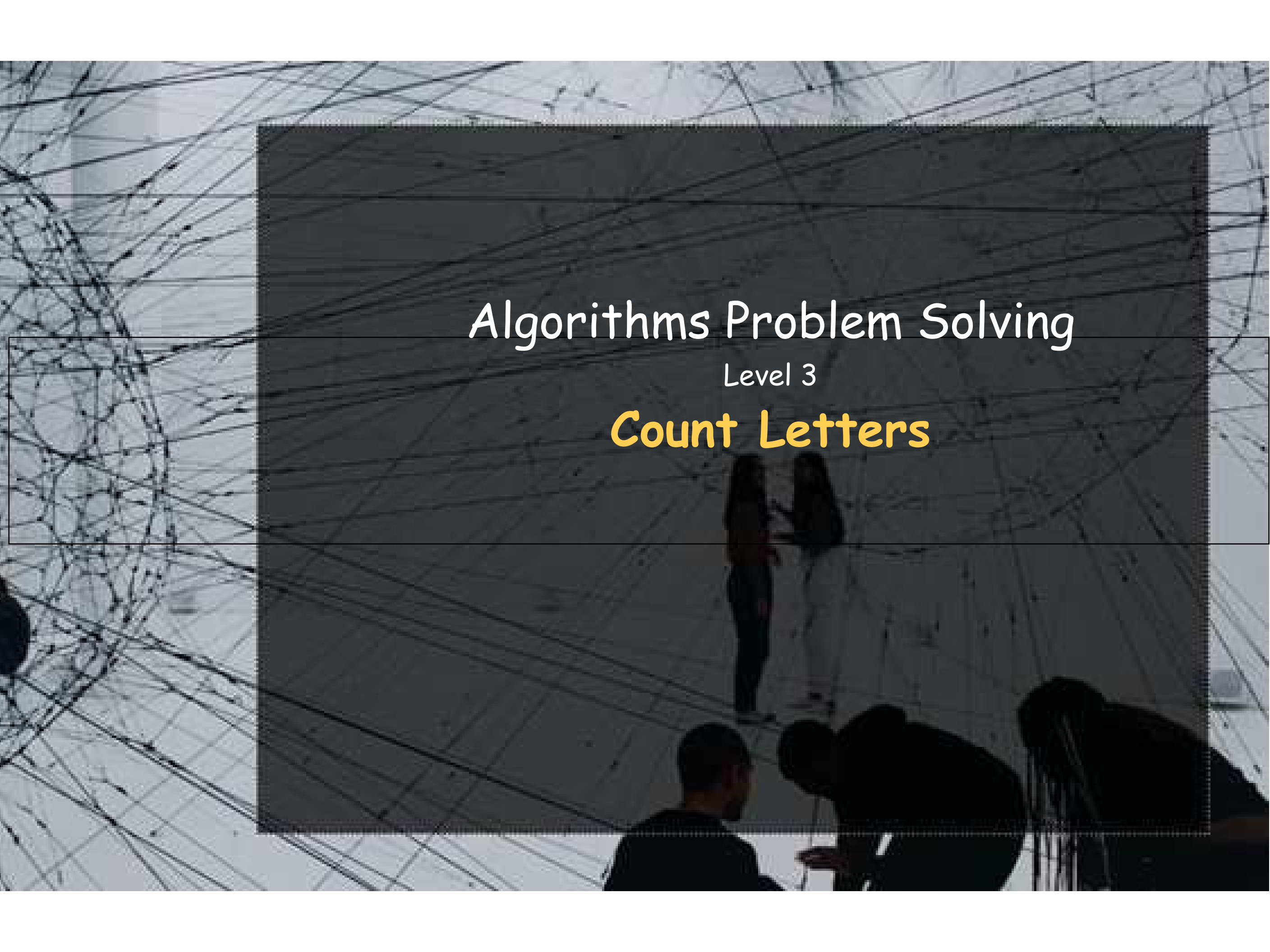


Algorithms Problem Solving

Level 3

Count Letters



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Definition

- ◆ This program reads a string from the user.
- ◆ It then reads a character and counts how many times that character appears in the string.
- ◆ The program checks every character in the string and increases the counter when a match is found.

Explanation

- ◆ ``ReadString()`` uses ``getline()`` to read the complete line, including spaces.
- ◆ ``ReadCharacter()`` reads the character that the user wants to search for.
- ◆ ``CountCharacter()`` receives the string and the character to search for.
- ◆ The ``for`` loop uses a range-based loop to examine every character in the string.
- ◆ ``current`` represents the character currently being checked.
- ◆ If ``current == ch``, the counter is increased by one.
- ◆ Finally, the function returns the total number of occurrences.

Procedures

- 1 Read the complete string from the user.
- 2 Read the character that should be counted.
- 3 Set `count` to `0`.
- 4 Loop through every character in the string.
- 5 Compare the current character with the requested character.
- 6 If they are equal, increase `count`.
- 7 Return the final count.
- 8 Print the number of occurrences.

💡 Example

Input string:

Sara Alterkawi

⚡ Character to count:

a

⚡ The character `a` appears four times in the string.

➡ Result

Letter 'a' Count = 4

Note

- ◆ ``getline(cin, input)`` is used instead of ``cin >> input`` because the string can contain spaces.
- ◆ The function ``CountCharacter()`` uses a range-based ``for`` loop:
`for (char current : str)`
- ◆ ``count`` starts from ``0`` because no characters have been matched initially.
- ◆ The comparison ``current == ch`` is case-sensitive.
- ◆ For example, ``a`` and ``A`` are treated as different characters.
- ◆ The parameter ``string& str`` is passed by reference, but the function does not modify the string. ``const string& str`` would be more appropriate here.

⊘ Warning

- ◆ The program counts only exact character matches.
- ◆ Searching for ``a`` does not count ``A``.
- ◆ The header `<sstream>` is included but is not used in this program.
- ◆ If the user enters more than one character when ``ReadCharacter()`` is called, ``cin >> ch`` reads only the first character.



Comparison

▨ `ReadString()`

- ✓ Reads a complete line of text.
- ✓ Uses `getline()`.
- ✓ Can read spaces inside the string.

▤ `ReadCharacter()`

- ✓ Reads a single character.
- ✓ Uses `cin >> ch`.
- ✓ Used to determine which character should be counted.

▨ `CountCharacter()`

- ✓ Searches through the string.
- ✓ Counts matching characters.
- ✓ Returns the final count.

▤ `ReadCharacter()`

- ✓ Gets the character to search for.
- ✓ Does not perform the counting.
- ✓ Its result is passed to `CountCharacter()`.

✦✦ Summary

- ✓ Read a complete string using ``getline()``.
- ✓ Read the character that should be counted.
- ✓ Loop through every character in the string.
- ✓ Compare each character with the requested character.
- ✓ Increase the counter when a match is found.
- ✓ Return and print the total number of occurrences.
- ✓ Character comparison is case-sensitive.

Thank you 😊